verify it hasn't been tampered with before processing the API request.

End-to-end security involves more than simply encrypting and decrypting data packets. Secure transport protocols, such as TLS, also ensure that both parties are adequately authenticated, and that data packets cannot be reordered or replayed. In the next few sections you'll see how to ensure the same protections are provided when using end-to-end security.

12.3.1 COSE

If you wanted to ensure end-to-end security of requests to a regular JSON-based REST API, you might be tempted to look at the JOSE (JSON Object Signing and Encryption) standards discussed in chapter 6. For IoT applications, JSON is often replaced by more efficient binary encodings that make better use of constrained memory and network bandwidth and that have compact software implementations. For example, numeric data such as sensor readings is typically encoded as decimal strings in JSON, with only 10 possible values for each byte, which is very wasteful compared to a packed binary encoding of the same data.

Several binary alternatives to JSON have become popular in recent years to overcome these problems. One popular choice is *Concise Binary Object Representation* (CBOR), which provides a compact binary format that roughly follows the same model as JSON, providing support for objects consisting of key-value fields, arrays, text and binary strings, and integer and floating-point numbers. Like JSON,